

# BILL OF MATERIALS

PROJECT REF:

547908

| Part/Item Identification |             |          |       |          |               |                                                        |                       |          | Design Stage |    |    | Mass (kg) |            |       |
|--------------------------|-------------|----------|-------|----------|---------------|--------------------------------------------------------|-----------------------|----------|--------------|----|----|-----------|------------|-------|
| Project                  | Bom Section | UID      | Issue | Revision | Type          | Part Description                                       | Part Number           | Quantity | Layout       | 3D | 2D | Density   | Mass Each  |       |
| 547908                   | 00          | 0000     | 01    | B        | Assembly      | Prototype Extractor Module Body                        | 547908-00-0000-01B    | 1        |              |    |    |           |            |       |
| 547909                   | 01          | 0001     | 01    | A        | Assembly      | Extractor Electronic Integration                       | 547909-01-0001-01A    | 1        |              |    |    |           |            |       |
| 547908                   | 02          | 0002     | 01    | A        | Assembly      | Video/Image Recording Setup                            | 547908-02-0002-01A    | 1        |              |    |    |           |            |       |
| 547908                   | 00          | 0003     | 01    | B        | Part          | Aluminium Extractor External Frame                     | 547908-00-0003-01B    | 1        |              |    |    | 2700      | 1.264      |       |
| 547908                   | 00          | 0004     | 01    | B        | Part          | PLA Extractor Shape Nozzle                             | 547908-00-0004-01B    | 5        |              |    |    | 1250      | 0.878      |       |
| 547908                   | 00          | 0005     | 01    | A        | Part          | PLA Locating Block                                     | 547908-00-0005-01A    | 2        |              |    |    | 1250      | 0.078      |       |
| 547908                   | 00          | 0006     | 01    | A        | Part          | Aluminium Mounting Bracket                             | 547908-00-0006-01A    | 2        |              |    |    | 2700      | 0.105      |       |
| 547908                   | 00          | 0007     | 01    | A        | Part          | Aluminium Extractor Vacuum Connector                   | 547908-00-0007-01A    | 1        |              |    |    | 1250      | 0.273      |       |
| 547908                   | 01          | 0008     | 01    | A        | Part          | PLA Electronics Housing                                | 547908-01-0008-01A    | 1        |              |    |    | 1250      | 0.086      |       |
| 547908                   | 01          | 0009     | 01    | A        | Part          | Alumnum Electronics Housing Bracket                    | 547908-01-0009-01A    | 2        |              |    |    | 2700      | 0.003      |       |
| 547908                   | 00          | 0010     | 01    | A        | Part          | Extractor Gasket                                       | 547908-00-0010-01A    | 1        |              |    |    | 1380      | 0.391      |       |
| 547908                   | 00          | 0011     | 01    | A        | Procurement   | 300 x 1200 mm PM2.5 HEPA Filter Paper with 20 mm Folds | 547908-00-0011-01A    | 1        |              |    |    |           | 0.040      |       |
| 547908                   | 01          | 0012     | 01    | A        | Procurement   | Sensiron Pressure sensor (+- 500Pa)                    | 547908-01-0012-01A    | 2        |              |    |    |           | 0.000      |       |
| 547908                   | 01          | 0013     | 01    | A        | Procurement   | 2.42 Inch SSD1309 128 x 64 OLED Display                | 547908-01-0013-01A    | 1        |              |    |    |           | 0.015      |       |
| 547908                   | 01          | 0014     | 01    | A        | Procurement   | RS PRO PVC 100m Hookup Wire Roll                       | 547908-01-0014-01A    | 0.01     |              |    |    |           | 0.060      |       |
| 547908                   | 01          | 0015     | 01    | A        | Procurement   | RS PRO Straight Through Hole Pin Headers               | 547908-01-0015-01A    | 1        |              |    |    |           | 0.022      |       |
| 547908                   | 01          | 0016     | 01    | A        | Procurement   | DHT11 Module (Temperature and Humidity sensor)         | 547908-01-0016-01A    | 1        |              |    |    |           | 1.000      |       |
| 547908                   | 01          | 0017     | 01    | A        | Procurement   | Arduino UNO Rev 3                                      | 547908-01-0017-01A    | 1        |              |    |    |           | 0.004      |       |
| 547908                   | 02          | 0018     | 01    | A        | Procurement   | 163cm Camera Tripod                                    | 547908-02-0018-01A    | 1        |              |    |    |           | 1.150      |       |
| 547908                   | 00          | 0019     | 01    | A        | Procurement   | RS PRO Polyvinyl Chloride Flexible Tubing 6mm OD       | 547908-00-0019-01A    | 1        |              |    |    | 1300      | 0.600      |       |
| 547908                   | 05          | 0020     | 01    | A        | Procurement   | 8 Channel Expansion Board                              | 547908-05-0020-01A    | 1        |              |    |    |           | 0.005      |       |
| 547908                   | 05          | 0021     | 01    | -        | Standard Part | Solderable Breadboard                                  | 547908-05-0021-01-    | 2        |              |    |    |           | 0.001      |       |
| 547908                   | 05          | 0022     | 01    | -        | Standard Part | USB A to B Cable                                       | 547908-05-0022-01-    | 1        |              |    |    |           | 0.100      |       |
| 547908                   | 05          | ISO 4762 | -     | -        | Standard Part | M4x20 Socket Cap Bolt                                  | 547908-05-ISO 4762--- | 12       |              |    |    |           | 7900       | 0.003 |
| 547908                   | 05          | ISO 4762 | -     | -        | Standard Part | M5x20 Socket Cap Bolt                                  | 547908-05-ISO 4762--- | 12       |              |    |    |           | 7900       | 0.004 |
| 547908                   | 05          | ISO 7089 | -     | -        | Standard Part | M4 Flat Washers                                        | 547908-05-ISO 7089--- | 12       |              |    |    |           | 7900       | 0.000 |
| 547908                   | 05          | ISO 7089 | -     | -        | Standard Part | M5 Flat Washers                                        | 547908-05-ISO 7089--- | 12       |              |    |    | 7900      | 0.000      |       |
| 547908                   | 05          | ISO 4762 | -     | -        | Standard Part | M10 Socket Cap Bolt                                    | 547908-05-ISO 4762--- | 4        |              |    |    | 7900      | 0.023      |       |
|                          |             |          |       |          |               |                                                        |                       |          |              |    |    |           | Total Mass |       |

|            | Cost Details (£)     |                       |            | Supply Chain Management |                                                                       |           |                                                           |  |
|------------|----------------------|-----------------------|------------|-------------------------|-----------------------------------------------------------------------|-----------|-----------------------------------------------------------|--|
| Mass Total | Cost Each (inc. VAT) | Cost Total (inc. VAT) | Company    | Supplier Part Number    | Hyperlink                                                             | Lead Time | Additional Info                                           |  |
|            |                      |                       | N/A        |                         |                                                                       |           |                                                           |  |
|            |                      |                       | N/A        |                         |                                                                       |           |                                                           |  |
|            |                      |                       | N/A        |                         |                                                                       |           |                                                           |  |
| 1.264      | £ 12.64              | £ 12.64               | Agfa       |                         |                                                                       |           | Material sponsored by Agfa, cost taken as online estimate |  |
| 4.390      | £ 0.88               | £ 4.39                | N/A        |                         |                                                                       |           | Printed in MechSpace                                      |  |
| 0.156      | £ 0.08               | £ 0.16                | N/A        |                         |                                                                       |           | Printed in MechSpace                                      |  |
| 0.210      | £ 0.11               | £ 0.21                | Agfa       |                         |                                                                       |           | Material sponsored by Agfa                                |  |
| 0.273      | £ 0.27               | £ 0.27                | Agfa       |                         |                                                                       |           | Material sponsored by Agfa                                |  |
| 0.086      | £ 0.09               | £ 0.09                | N/A        |                         |                                                                       |           | Printed in MechSpace                                      |  |
| 0.005      | £ 0.00               | £ 0.01                | Agfa       |                         |                                                                       |           | Material sponsored by Agfa                                |  |
| 0.391      | £ 0.78               | £ 0.78                | Agfa       |                         |                                                                       |           | Material sponsored by Agfa                                |  |
| 0.040      | £ 1.24               | £ 1.24                | AliExpress |                         | <a href="https://www.aliexpress.com/">https://www.aliexpress.com/</a> | 2W        |                                                           |  |
| 0.000      | £ 22.04              | £ 44.09               | RS         | 133-1913                | <a href="https://uk.rs-online.com/">https://uk.rs-online.com/</a>     | 3D        |                                                           |  |
| 0.015      | £ 12.99              | £ 12.99               | Amazon     |                         | <a href="https://www.amazon.co.uk/">https://www.amazon.co.uk/</a>     | 1D        |                                                           |  |
| 0.001      | £ 39.30              | £ 0.39                | RS         | 183-9225                | <a href="https://uk.rs-online.com/">https://uk.rs-online.com/</a>     | 3D        |                                                           |  |
| 0.022      | £ 7.58               | £ 7.58                | RS         | 251-8632                | <a href="https://uk.rs-online.com/">https://uk.rs-online.com/</a>     | 3D        |                                                           |  |
| 1.000      | £ 7.12               | £ 7.12                | RS         | 216-3753                | <a href="https://uk.rs-online.com/">https://uk.rs-online.com/</a>     | 3D        |                                                           |  |
| 0.004      | £ 22.34              | £ 22.34               | RS         | 715-4081                | <a href="https://uk.rs-online.com/">https://uk.rs-online.com/</a>     | 3D        |                                                           |  |
| 1.150      | £ -                  |                       | Amazon     |                         | <a href="https://www.amazon.co.uk/">https://www.amazon.co.uk/</a>     | 4D        |                                                           |  |
| 0.600      | £ 12.06              | £ 12.06               | Agfa       | 336-821                 | <a href="https://uk.rs-online.com/">https://uk.rs-online.com/</a>     | 3D        | Material sponsored by Agfa                                |  |
| 0.005      | £ 3.99               | £ 3.99                | Amazon     |                         |                                                                       |           |                                                           |  |
| 0.002      | £ 1.32               | £ 2.64                | RS         | 286-2016                | <a href="https://uk.rs-online.com/">https://uk.rs-online.com/</a>     | 3D        |                                                           |  |
| 0.100      | £ 3.35               | £ 3.35                | RS         | 236-9076                | <a href="https://uk.rs-online.com/">https://uk.rs-online.com/</a>     | 3D        |                                                           |  |
| 0.032      | £ 0.04               | £ 0.48                | N/A        |                         |                                                                       |           | Supplied from MechSpace                                   |  |
| 0.048      | £ 0.30               | £ 3.60                | N/A        |                         |                                                                       |           | Supplied from MechSpace                                   |  |
| 0.004      | £ 0.02               | £ 0.24                | N/A        |                         |                                                                       |           | Supplied from MechSpace                                   |  |
| 0.005      | £ 0.02               | £ 0.24                | N/A        |                         |                                                                       |           | Supplied from MechSpace                                   |  |
| 0.092      | £ 0.24               | £ 0.96                | N/A        |                         |                                                                       |           | Supplied by Agfa                                          |  |
| 9.895      | Total Cost           | £ 141.85              |            |                         |                                                                       |           |                                                           |  |